

traffic congestion. Passengers may not realise that their flight has been delayed and miss the boarding time.

Slow boarding: The boarding process can take a long time, especially if passenger's do not follow the airline's instructions. This can lead to delays, missed connections, and other problems.

Security issues: Passengers may have problems with security, such as not being able to bring certain items on board or having to go through additional screening. This can cause delays.

Communication issues: Airlines may not provide clear instructions or communicate effectively with passengers during the boarding process. Last minute gate changes can lead to confusion and delays.

Hassle-free flight boarding experience with AI and ML

AI and ML can help resolve most of the above problems. Let's use the same anecdote to see how.

Jayashree had been eagerly waiting to visit her family in India for months. She arrived at Dubai airport four hours before her flight, checked-in her luggage and completed the immigration and security checks. Since she had several hours before her flight, she decided to do some shopping and grab some food. However, she lost track of time and didn't know that she should reach the boarding gate 30 minutes before the scheduled departure of the flight.



Image courtesy: Oleksandr Pidvalnyi

Fortunately, the airport had an AI-powered boarding system that ensured seamless boarding for all passengers. The system used IoT and machine learning to keep track of passengers' movements in the airport and estimate their time of arrival at the gate.

When Jayashree was predicted late, the system immediately alerted the gate agent and airline staff to locate her. The system used facial recognition and other tracking technologies to pinpoint her location in the airport, and within minutes, a staff member was able to find her and escort her to the gate.

Jayashree was able to board her flight without any hassle and also enjoy her shopping experience — all thanks to the seamless and efficient boarding process facilitated by AI, IoT, and machine learning.

AI, ML, data analytics and IoT services provided by AWS

The hassle-free flight boarding experience described above can happen easily by leveraging AWS cloud services.

There are a lot of IoT devices available to track the movements of a passenger/airline staff using geofencing and BLE (Bluetooth low energy). Beacons, for example, can cover a range of 5m to 2.5km. They emit signals to devices like mobile phones, and are mapped with their location. This information is stored in AWS RDS, which helps to detect the location of the passenger. Motion cameras can also be used.

AWS provides a range of AI, ML, data analytics, and IoT (Internet of Things) services that can be used to build intelligent and data-driven applications. Some of these key services are listed in Table 1.

AI SERVICES	Machine Learning services	Data Analytics services	IoT Services
Amazon Rekognition	Amazon SageMaker	Amazon Athena	AWS IoT Core
Amazon Lex	Amazon Forecast	Amazon EMR	AWS Greengrass
Amazon Comprehend	Amazon Fraud Detector	Amazon Kinesis	AWS IoT Analytics
Amazon Translate	Amazon CodeGuru	Amazon Redshift	AWS IoT Device Defender
Amazon Polly	Amazon Elastic Inference	Amazon QuickSight	AWS IoT Device Management
Amazon Transcribe	AWS Deep Learning AMIs	AWS Glue	Amazon FreeRTOS
Amazon Textract	AWS Deep Learning Containers	AWS Data Pipeline	AWS IoT 1-Click
Amazon Personalize	AWS DeepLens	AWS Lake Formation	AWS IoT Events
Amazon Kendra	AWS DeepRacer	Amazon MSK	AWS IoT SiteWise
Amazon Lookout for Vision	AWS DeepComposer	Amazon Elasticsearch Service	AWS IoT Things Graph
Amazon Lookout for Equipment		AWS Cassandra Service (MCS)	
Amazon Augmented AI (Amazon A2I)		AWS Data Exchange	
Alexa			

Table 1: Key web services offered by Amazon